

Question:

A baby is born after the 32nd gestational week by cesarean delivery. The mother suffered from gestational diabetes; however, she had no other pregnancy-related diseases and was otherwise healthy. The baby has a blood pressure of 100/58 mm Hg, heart rate of 104/min, and oxygen saturation of 88%. The child has tachypnea, subcostal and intercostal retractions, nasal flaring, and cyanosis. The cyanosis is responding well to initial administration of oxygen. The nasogastric tube was positioned without problems. Which of the following is the most likely diagnosis?

Answer Choices:

- A. Neonatal respiratory distress syndrome (NRDS)
- B. Tracheoesophageal fistula
- C. Pneumonia
- D. Congenital heart anomaly with right-to-left shunt

Reasoning + Answer:

<think>

Finding Reasoning Paths :

1. **Tachypnea, subcostal retractions, intercostal retractions, nasal flaring, and cyanosis** suggest respiratory distress, which could be due to **neonatal respiratory distress syndrome (NRDS)** or **pneumothorax**.
2. **Prematurity** (born at 32 weeks) is a significant risk factor for **neonatal respiratory distress syndrome**, caused by insufficient surfactant production in premature infants.
3. **Gestational diabetes** may contribute to metabolic abnormalities or infections, but there is no direct evidence linking it to the respiratory findings.
4. **Abnormal oxygen saturation** (88%) and cyanosis indicate impaired gas exchange, which could result from **respiratory distress syndrome** or **pneumothorax**.
5. **Examination of the nasogastric tube** rules out esophageal obstruction or feeding problems, leaving the respiratory system as the primary focus.

Reasoning Process :

1. **Prematurity and Surfactant Deficiency**:
The infant was born at 32 weeks gestation, which is considered preterm. Preterm infants are at high risk of **neonatal respiratory distress syndrome (NRDS)** due to insufficient surfactant production. Surfactant deficiency leads to alveolar collapse, resulting in tachypnea, hypoxemia (abnormal oxygen saturation), and respiratory distress symptoms such as subcostal retractions, intercostal retractions, nasal flaring, and cyanosis. The improvement in oxygen saturation with oxygen administration is consistent with NRDS.
2. **Pneumothorax**:
Pneumothorax is another possible cause of respiratory distress in preterm infants, but it is not explained by the nasogastric tube or other findings. Pneumothorax typically presents with a fixed heart rate and bounding pulses, which are not described here. While there is no evidence of pneumothorax in this case, it is a potential alternative to consider.
3. **Gestational Diabetes and Infections**:
Gestational diabetes is a risk factor for maternal infections, which can lead to neonatal infections or metabolic disorders. However, there is no indication in the question that the infant has a fever, abnormal physical findings, or other signs of infection. Additionally, gestational diabetes is not directly associated with the respiratory findings described.
4. **Other Causes of Respiratory Distress**:
Conditions such as pneumonia or congenital heart defects could cause respiratory distress, but these are not supported by the clinical findings or examination results. Congenital heart defects are unlikely given the absence of murmurs, cyanosis without bluing of the lips, or other heart-related symptoms. Pneumonia would typically present with productive cough, rhonchi, or auscultatory findings, none of which are described here.

Conclusion :

The most likely diagnosis is **neonatal respiratory distress syndrome (NRDS)**, caused by surfactant deficiency due to premature birth at 32 weeks. The improvement in oxygen saturation with oxygen administration further supports this diagnosis.

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<answer>

Neonatal respiratory distress syndrome (NRDS)

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